



NARAYANA
IIT ACADEMY
INDIA

41
YEARS
OF EXCELLENCE

JEE ADVANCED GRAND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA

45
YEARS
OF EXCELLENCE

Sec: SR.IIT_*CO-SC
Time: 3HRS

Date: 17-03-24
Max. Marks: 186

Name of the Student: _____

H.T. NO:

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14-04-24_SR.STAR CO-SUPER CHAINA_JEE-ADV_GTA-5(P2)_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

Time: 3 Hrs**IMPORTANT INSTRUCTIONS****Max Marks: 186****MATHS:**

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 6)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 7 – 14)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 15 – 18)	Questions with Comprehension Type (2 Comprehension-2+2=4Q)	3	0	4	12
Total				18	62

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 24)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 25 – 32)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 33 – 36)	Questions with Comprehension Type (2 Comprehension-2+2=4Q)	3	0	4	12
Total				18	62

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 42)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 43 – 50)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 51 – 54)	Questions with Comprehension Type (2 Comprehension-2+2=4Q)	3	0	4	12
Total				18	62

THE NARAYANA GROUP

MATHEMATICS

Max. Marks: 62

SECTION-I

(Single Correct Answer Type)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

1. Given that the real numbers, s, t satisfy $19s^2 + 99s + 1 = 0$, $t^2 + 99t + 19 = 0$ and $st \neq 1$.

The absolute value of $\frac{st + 4s + 1}{t}$ is

A) Divisible by 99

B) Divisible by 19

C) Divisible by 5

D) Divisible by 2

2. For all $x \geq 1$, let $f(x)$ be the function defined as

$$f(x) = \begin{cases} [x] \cdot \left| x - [x] - \frac{1}{2[x]} \right|, & \text{if } x < [x] + \frac{1}{[x]} \\ f\left(x - \frac{1}{[x]}\right) & \text{otherwise} \end{cases} \quad (\text{where } [.] \text{ denote greatest integer function})$$

The number of points at which $f(x)$ is not differentiable in the interval $x \in (4, 5)$ is

A) 8

B) 7

C) 4

D) 2

3. If $y = y(x)$ satisfies the differential equation

$$8\sqrt{x}(\sqrt{9+\sqrt{x}})dy = \left(\sqrt{4+\sqrt{9+\sqrt{x}}}\right)^{-1} dx, x > 0 \text{ \& } y(0) = \sqrt{7}, \text{ then the value of } y(256) =$$

A) 16

B) 9

C) 3

D) 1

4. The absolute value of $\frac{\int_0^{\pi/2} (x \cos x + 1)e^{\sin x} dx}{\int_0^{\pi/2} (x \sin x - 1)e^{\cos x} dx}$ is equal to k then the value of $[k]$

where $[.]$ denotes G I F

A) 0

B) 1

C) 3

D) 2

5. In the complex plane, let A be the set of solutions to the equation $z^3 - 8 = 0$ and let B be the set of solutions to the equation $z^3 - 8z^2 - 8z + 64 = 0$. Then the greatest distance between a point of A and a point of B is $\lambda\sqrt{21}$, where $\lambda =$

A) 2

B) 1

C) 3

D) 4

6. The number of arrangements of the letters of the word PROBLEMATIC (sixth letter) taking all such that neither starting with P, nor ending with C, nor M in the middle is

A) $8!$ (746)B) $8!$ (373)C) $9!$ (82)D) $10!$ (23)

SECTION-II

(Multiple Correct Answer Type)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases.

7. If two circles touch the parabola $75y^2 = 64(5x - 3)$ at $(\frac{6}{5}, \frac{8}{5})$ and the x-axis then
- A) centre of the smaller circle is (2,1)
- B) radius of the bigger circle is 4
- C) equation of smaller circle is $x^2 + y^2 - 4x - 2y + 4 = 0$
- D) radius of the smaller circle is 2
8. If $S = \tan^{-1}\left(\frac{4}{7}\right) + \tan^{-1}\left(\frac{4}{19}\right) + \tan^{-1}\left(\frac{4}{39}\right) + \tan^{-1}\left(\frac{4}{67}\right) + \dots = T_1 + T_2 + T_3 + \dots$ then
- A) sum of first 11 terms of series = $\tan^{-1}\left(\frac{48}{29}\right)$
- B) sum of first 11 terms = $\tan^{-1}\left(\frac{44}{27}\right)$
- C) sum of infinite terms of series = $\frac{\pi}{2}$
- D) sum of infinite terms of series = $\tan^{-1}(2)$
9. Let $p(x) = 1 - 2x + 3x^2 - 4x^3$ & $f(x)$ is a polynomial function satisfying $f(0) = -19$ and $f(x) - f'(x) = p(x)$ then which of the following is/are true
- A) $f(x) = 0$ has only one negative real root
- B) $f(x) = 0$ has only one positive real root
- C) The equation $f(x) = p(x)$ has no real root
- D) The equation $f(x) = p(x)$ has two real roots
10. The value of the determinant $\begin{vmatrix} \sin(x+A) & \cos(x+A) & a+x\sin A \\ \sin(x+B) & \cos(x+B) & b+x\sin B \\ \sin(x+C) & \cos(x+C) & c+x\sin C \end{vmatrix}$ is independent of
- A) x only
- B) x, A, B and C only
- C) x, a, b and c only
- D) x, a, b, c, A, B and C

- D) The probability of winning Game A is $\frac{4}{81}$ greater than the probability of winning Game B

- D) If the planes $\vec{r} \cdot \vec{n}_1 = d_1, \vec{r} \cdot \vec{n}_2 = d_2, \vec{r} \cdot \vec{n}_3 = d_3$ have a common line of intersection then it is not necessary that $d_1(\vec{n}_2 \times \vec{n}_3) + d_2(\vec{n}_3 \times \vec{n}_1) + d_3(\vec{n}_1 \times \vec{n}_2) = \vec{0}$

- D) $f(2011) = f(2012)$

14. Let $S \equiv \frac{x^2}{4} + \frac{y^2}{1} = 1$ be a given ellipse. C_1, C_2 are two circles touching the ellipse, at end points of minor axis, C_1 at $A(0,1)$ and C_2 at $B(0,-1)$. If radius of C_1 is unity, and a line L which is normal to C_1 , is tangent to curves S and C_2 then which of the following options is/are correct

A) Slope of L can be $-\frac{\sqrt{3}}{2}$

B) Equation of L can be $\sqrt{2}x + 3y - 6 = 0$

C) Centre of C_2 can be $(0, -5 - 2\sqrt{7})$

D) Radius of C_2 can be $4 + 2\sqrt{7}$

SECTION-III (Paragraph Type)

This section contains 2 groups of question. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choice (A), (B), (C) AND (D) for its answer, out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph for Questions 15 and 16:

A curve $y = f(x)$ passes through the point $P(1,1)$. The normal to the curve at P is $a(y-1) + (x-1) = 0$. If the slope of the tangent at any point on the curve is proportional to the ordinate of the point, then:

15. The equation of the curve $y = f(x)$ is

A) $y = e^{a(x-1)}$

B) $y = e^{a(x^2-1)}$

C) $y = e^{a(1-|x|)}$

D) $y = e^{a(|x|-1)}$

16. The area bounded by the curve $y = f(x)$, y -axis and the normal to the curve at P is:

A) $\frac{1}{a} \left(a - \frac{1}{2} + e^{-a} \right)$

B) $\frac{1}{a} \left(a + \frac{1}{2} + e^{-a} \right)$

C) $\frac{1}{a} \left(a - \frac{1}{2} + e^a \right)$

D) $\frac{1}{a} \left(a + \frac{1}{2} + e^a \right)$

Paragraph for Questions 17 and 18:

Consider the lines L_1 and L_2 defined by $L_1 : x\sqrt{2} + y - 1 = 0$ and $L_2 : x\sqrt{2} - y + 1 = 0$. For a fixed constant λ , let C be the locus of a point P such that the product of the distance of P from L_1 and the distance of P from L_2 is λ^2 . The line $y = 2x + 1$ meets C at two points R and S , where the distance between R and S is $\sqrt{360}$.

Let the perpendicular bisector of RS meet C at two distinct points R' and S' . Let D be the square of the distance between R' and S'

17. The value of λ^2 is _____

A) 25

B) 12

C) 36

D) 19

18. The value of $\frac{D}{5}$ is _____

A) 20.57

B) 6

C) 7.5

D) 8

PHYSICS

Max. Marks: 62

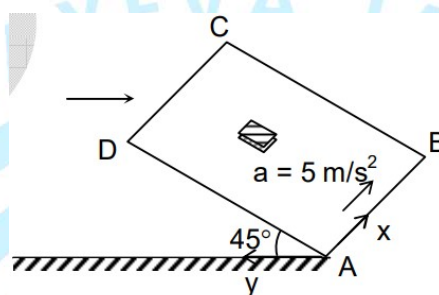
SECTION-I

(Single Correct Answer Type)

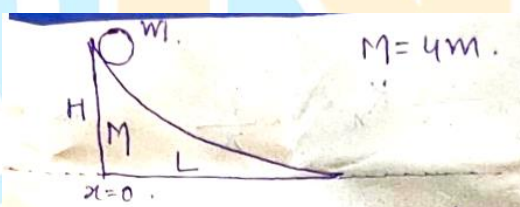
This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

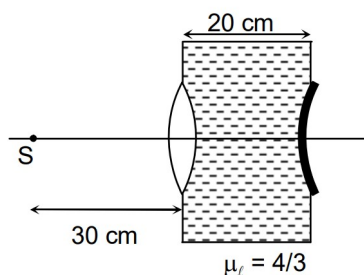
19. An inclined plane ABCD is shown in the figure. It is inclined at an angle of 45° from horizontal. A block is placed on the inclined plane. The plane is accelerating along x-axis with an acceleration, $a = 5 \text{ m/s}^2$. The minimum value of coefficient of friction between the block and the plane required, so that the block does not slip on the plane, is ($g = 10 \text{ m/s}^2$)



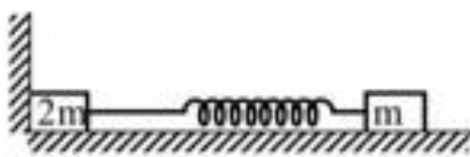
- A) $\sqrt{1.5}$ B) 1 C) 2 D) $\sqrt{2}$
20. A solid sphere of mass 'm' and Radius ($R \ll H$) is released from rest from top of wedge (mass=M) as shown. Assuming sphere doesn't slip over wedge curved surface, it rolls down. The speed of sphere just before it hits the ground is



- A) $\sqrt{\frac{16gH}{15}}$ B) $\sqrt{\frac{40gH}{33}}$ C) $\sqrt{2gH}$ D) $\frac{4}{5}\sqrt{2gH}$
21. A thin equiconvex lens of glass (refractive index $\mu_g = 3/2$) having focal length 20 cm in air, is placed in front of a convex mirror of radius of curvature $R_0 = 40 \text{ cm}$. The optical axis of the lens and the mirror are coinciding. The separation between the lens and the mirror is $d = 20 \text{ cm}$ and a transparent liquid (refractive index $\mu_l = \frac{4}{3}$) is filled between the lens and the mirror. A point source S of light is placed at a distance 30 cm from the lens and it is on the optical axis of the system as shown in the figure. Then, choose the correct option.



- A) Final image is formed at the source S itself
 B) Final image is formed at the pole of the mirror
 C) Final image is formed 10 cm behind the lens
 D) Final image is formed at infinity
22. An uniformly charged solid sphere of charge Q and radius R is placed in a vacuum. The electrostatic energy stored in a cone of semi-angle θ and slant height ' $2R$ ' with the vertex of the cone lying at the centre of sphere is x . Then for small ' θ ' value of x is
- A) $\frac{7Q^2 \sin^2 \frac{\theta}{2}}{80\pi \epsilon_0 R}$ B) $\frac{7Q^2 \sin^2 \frac{\theta}{2}}{160\pi \epsilon_0 R}$ C) $\frac{Q^2 \sin^2 \frac{\theta}{2}}{16\pi \epsilon_0 R}$ D) $\frac{Q^2 \sin^2 \frac{\theta}{2}}{8\pi \epsilon_0 R}$
23. A ideal gas whose adiabatic exponent equals γ is expanded so that the amount of heat transferred to the gas is equal to twice of decrease of its internal energy. The equation of the process is $TV^{\frac{\gamma-1}{K}} = \text{constant}$ (where T and V are absolute temperature and volume respectively). Then the value of ' K ' is.
- A) 1 B) 2 C) 3 D) 4
24. Two blocks of mass m and $2m$ connected by a weightless spring of stiffness k rest on a smooth horizontal plane. Block of mass m is shifted to a small distance x to the left and then released. If the duration between releasing mass m and breaking off of mass $2m$ is Δt , then average force (in newton) exerted by wall on mass $2m$ is given as $\frac{x}{\Delta t} \sqrt{mkn}$. Find the value of n .



- A) 1 B) 2 C) 3 D) 4

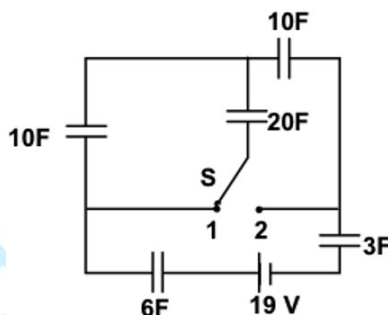
SECTION-II

(Multiple Correct Answer Type)

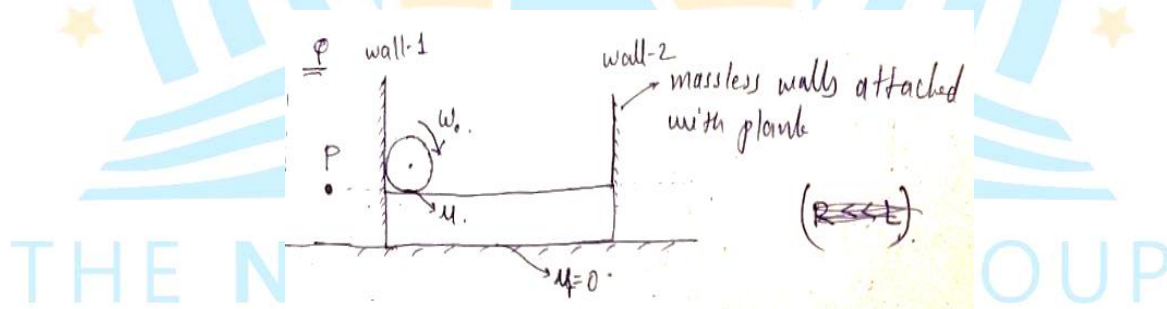
This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases.

25. When the steady state is reached by keeping switch in position - 1, then switch 'S' is transferred from position 1 to 2. Then

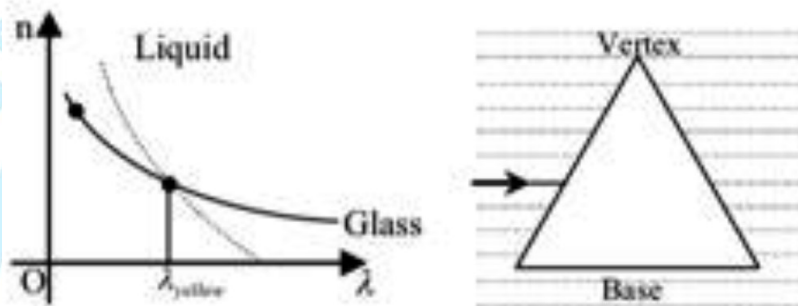


- A) Charge on 3F capacitor is 30 C after S is transferred from position 1 to 2.
 B) Charge on 3F capacitor is $\frac{650}{19}$ C after S is transferred from 19 position 1 to 2.
 C) Work done by the battery is 380 Joule after S is transferred from position 1 and 2.
 D) Work done by the battery is 80 Joule after S is transferred from position 1 and 2.
26. On a plank of mass 'm' and length-L, a ring (m,R) rotated by ω_0 , is kept gently on left end of plank initially plank is at rest on smooth ground. It is observed that ring attain pure rolling just before hitting wall-2 elastically (point P lies at same level as of top surface of the plank)

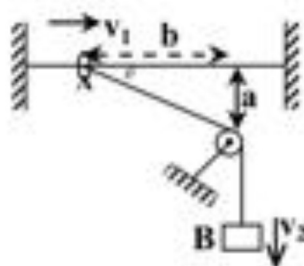


- A) $\frac{\omega_0 R}{3} = \sqrt{\mu g L}$
 B) $\frac{\omega_0 R}{3} = \sqrt{2\mu g L}$
 C) After colliding elastically with wall-2 for first time, the ring attain pure rolling before colliding wall-1
 D) Angular momentum of the system about point P will remain conserved though out the motion of the system

27. A standing wave is setup in a string fixed at both ends. Then
- A) The sum of total energy per unit length at nodes and antinodes is constant
 - B) The total energy of a point midway between a node and an adjacent antinode is constant
 - C) If the string has a standing wave and a component of travelling wave, then the kinetic energy of the points of minimum amplitude is non-zero at some instant of time.
 - D) If the string has a standing wave and a component of travelling wave, then the potential energy of the points of maximum amplitude is non zero at some instant of time.
28. A glass prism is immersed in a hypothetical liquid. The curves showing the refractive index n as a function of wavelength λ for glass and liquid are as shown in the figure. When a ray of white light is incident on the prism parallel to the base:



- A) Yellow ray travels without deviation
 - B) Blue ray is deviated towards the vertex
 - C) Red ray is deviated towards the base
 - D) There is no dispersion
29. Ring A can slide on a smooth horizontal rod and block B can move vertically. Both are attached by an inextensible string. At a certain instant velocity of ring is v_1 and of block is v_2 . At that instant ring is at a horizontal distance b from pulley as shown. Then



A) $v_2 = v_1 \cos \theta$

B) $a_2 = a_1 \cos \theta$, where a_2 and a_1 represent the accelerations of block and ring respectively at that instant

C) $v_1 = v_2 \cos \theta$

D) Angular velocity of the ring with respect to an observer on the pulley is $\frac{v_1 \sin \theta}{\sqrt{a^2 + b^2}}$

30. In a photoelectric effect experiment, the frequency of incident radiation is increased to n times ($n > 1$). Then

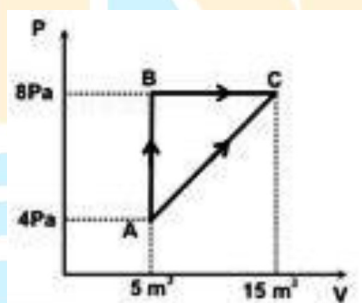
A) Saturation current remains constant

B) Stopping potential becomes more than n times of initial stopping potential

C) The difference of the two stopping potentials is equal to $(n - 1) \left(V + \frac{\phi}{e} \right)$, where f is the work function and V is the initial stopping potential

D) The maximum kinetic energy of the photoelectrons increases.

31. In the given graph, an ideal gas can change its state from A to C by two paths ABC or AC. Then



A) If the internal energy of gas at 'A' is 10 J and the amount of heat supplied in path AC is 200J, then the internal energy of gas at 'C' is 150 J

B) If the internal energy of gas at 'A' is 10 J and the amount of heat supplied in path AC is 200J, then the internal energy of gas at 'C' is 75 J

C) If the internal energy of gas at state B is 20 J. Then the amount of heat supplied to the gas to go from A to B may be 5 Joule

D) If the internal energy of gas at state B is 20J. Then the amount of heat supplied to the gas to go from A to B must be 10 Joule

32. An infinite sheet having charge density $\sigma = \frac{-a}{(b^2 + r^2)^{3/2}}$, where 'a' and 'b' are positive constant and r is the radial distance from origin point – 0. Then (origin lies on the plate)
- A) E in front of the plate at very small distance from origin is $\frac{a}{2b^3 \epsilon_0}$
- B) E at any point in space is a central field directed towards the origin
- C) E in front of plate at distance 'b' (measured normally) from origin is $\frac{a}{8b^3 \epsilon_0}$
- D) The E is conservative in nature (considering the region of only one side of plate)

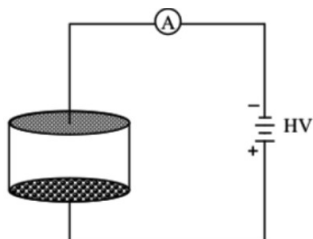
SECTION-III (Paragraph Type)

This section contains 2 groups of question. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choice (A), (B), (C) AND (D) for its answer, out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph for Questions 33 and 34:

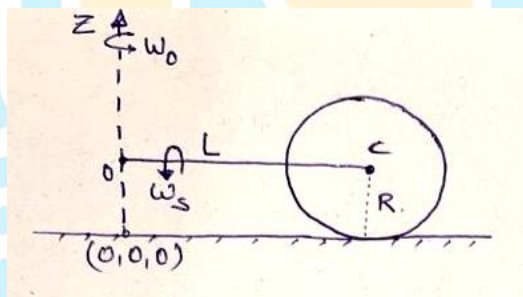
Consider an evacuated cylindrical chamber of height h having rigid conducting plates at the ends and an insulating curved surface as shown in the figure. A number of spherical balls made of a light weight and soft material and coated with a conducting material are placed on the bottom plate. The balls have a radius $r \ll h$. Now a high voltage source (HV) is connected across the conducting plates such that the bottom plate is at $+V_0$ and the top plate at $-V_0$. Due to their conducting surface, the balls will get charged, will become equipotential with the plate and are repelled by it. The balls will eventually collide with the top plate, where the coefficient of restitution can be taken to be zero due to the soft nature of the material of the balls. The electric field in the chamber can be considered to be that of a parallel plate capacitor. Assume that there are no collisions between the balls and the interaction between them is negligible. (Ignore gravity)



33. Which one of the following statements is correct?
- A) The balls will bounce back to the bottom plate carrying the opposite charge they went up with
- B) The balls will execute simple harmonic motion between the two plates
- C) The balls will bounce back to the bottom plate carrying the same charge they went up with
- D) The balls will stick to the top plate and remain there
34. The average current in the steady state registered by the ammeter in the circuit will be
- A) Proportional to $V_0^{1/2}$ B) Proportional to V_0^2
- C) Proportional to the potential V_0 D) Zero

Paragraph for Questions 35 and 36:

A solid sphere of mass 'M' and radius 'R' is moving on X-Y plane with centre 'C' rotating about Z-axis in circular path (centred at 'O'). A shaft OC of length 'L', as shown is free to rotate about Z-axis with ends fixed at 'O' and 'C'. The sphere is rotating about shaft 'OC' with ' ω_s ' and the shaft about Z-axis ω_0 . The sphere doesn't slip on the X-Y plan.



35. Total kinetic energy of the shell is
- A) $\frac{1}{2}M(\omega_0 L)^2 + \frac{1}{3}MR^2(\omega_s^2 + \omega_0^2)$ B) $\frac{1}{2}M(\omega_0 L)^2 + \frac{1}{3}MR^2\omega_s^2$
- C) $\frac{1}{2}M(\omega_0 L)^2 + \frac{1}{5}MR^2(\omega_s^2 + \omega_0^2)$ D) $\frac{1}{2}M(\omega_0 L)^2 + \frac{1}{5}MR^2\omega_s^2$
36. The number of points on the shell at any instant whose speed is zero
- A) 0 B) 1 C) 2 D) Infinite

CHEMISTRY

Max. Marks: 62

SECTION-I

(Single Correct Answer Type)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

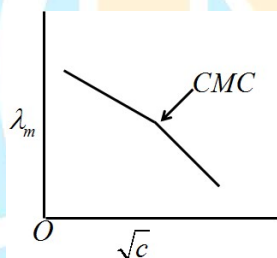
37. The EMF of the cell $Ag|AgCl, 0.2M KCl||0.1M AgNO_3|Ag$ is 0.45. The solubility product of $AgCl$ at $25^\circ C$ is _____

[Given $\frac{2.303RT}{F} = 0.06$, Antilog (0.5) = 3.16] (Assuming 100% ionization electrolytes).

- A) $6.3 \times 10^{-12} M^2$ B) $12.6 \times 10^{-10} M^2$ C) $6.3 \times 10^{-10} M^2$ D) $3.15 \times 10^{-9} M^2$

38. Which of the following statements is **incorrect**?

- A) With respect to adsorption: $\Delta G < 0$; $\Delta S < 0$; $\Delta H < 0$.
- B) When the precipitate of SnO_2 is peptised with NaOH solution, a positively charged sol is obtained.
- C) Variation of molar conductivity (λ_m) with concentration of sodium stearate in solution



can be represented as

- D) The coagulating power of electrolytes having ions $Na^\oplus$, Al^{3+} and Ba^{2+} for Arsenic sulphide sol increases in the order: $Na^\oplus < Ba^{2+} < Al^{3+}$

39. Bromine halide (**R**) + $H_2O \rightarrow X + Y$, [major products]

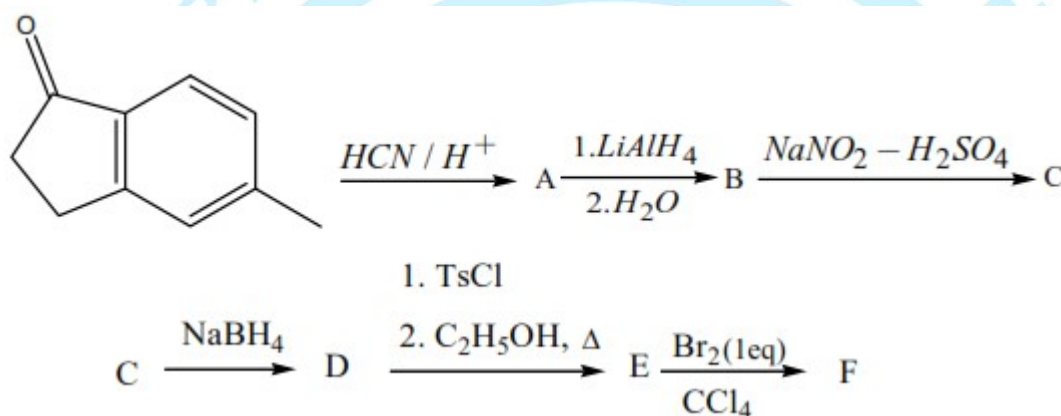
Electronic geometry of **R** is a distorted octahedron. Choose correct pair of X and Y .

- A) X = HBr Y = HOF
- B) X = HBrO Y = HF
- C) Y = HBrO₂ X = HF
- D) Y = HBrO₃ X = HF

40. Which of the following statement(s) is incorrect?

- A) The geometries of the ammonia complexes of Ni^{2+} , Pt^{2+} & Zn^{2+} , respectively are Octahedral, square planar & tetrahedral.
- B) $[Ni(CN)_4]^{2-}$, $[NiCl_4]^{2-}$ and $[Ni(CO)_4]$ are all paramagnetic
- C) In $Cr_2O_7^{2-}$, all 'chromium – oxygen' bond lengths are not identical
- D) $[(Ph_3P)_3RhCl]$ is known as Wilkinson catalyst and is used for the hydrogenation of alkenes

41. In the given reaction sequence, the compound F (major) is



The compound F (major) is.

- A)
- B)
- C)
- D)

42. K_2MnO_4 in acidic medium changes from green to purple after some time. This change is NOT accompanied by
- A) Disproportionation of K_2MnO_4
- B) Decrease in pH of solution
- C) Separation of brown mass from the solution
- D) Formation of tetrahedral oxoanion of Mn with oxidation state +7

SECTION-II

(Multiple Correct Answer Type)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases.

43. Which of the following statement(s) is/are true ?
- A) In N_2 , the doubly degenerate π_{2p} orbitals are completely filled .
- B) In O_2 , the energy of σ_{2p_z} orbital is lower than the doubly degenerate π_{2p} orbitals .
- C) In transformations $N_2 \rightarrow N_2^{2-}$, both HOMO and magnetic behavior changes.
- D) A π_{2p}^* orbital has two nodal planes.
44. If NaCl crystal has two 'F' centers per unit cell. Then the possible formula per unit cell
- A) $Na_4Cl_{3.75}$ B) Na_4Cl_3 C) $Na_4Cl_{3.375}$ D) $Na_4Cl_{3.5}$
45. Which of the following statement(s) is/are correct statement(s)?
- A) $Pb^{2+} \xrightarrow{\text{dilute HCl}} PbCl_2 \downarrow \xrightarrow{\text{conc KCl}} [PbCl_4]^{2-}$
(White)
- B) $Pb^{2+} \xrightarrow{NaOH} Pb(OH)_2 \downarrow \xrightarrow{NaOH} [Pb(OH)_4]^{2-}$
(White)
- C) $Pb^{2+} \xrightarrow{\text{dilute H}_2\text{SO}_4} PbSO_4 \downarrow \xrightarrow[\text{ammonium acetate in the presence of } NH_3]{6M \text{ Solution of}} [Pb(CH_3COO)_4]^{2-}$
(White)
- D) $Pb^{2+} \xrightarrow{\text{aqueous KI}} PbI_2 \downarrow \xrightarrow[\text{solution of KI}]{\text{excess of more concentrated (6M)}} [PbI_4]^{2-}$
(yellow)

46. Which among the following statement(s) is(are) true for the extraction of aluminium from bauxite?

- A) Hydrated Al_2O_3 precipitates, when CO_2 is bubbled through a solution of sodium aluminate.
- B) Addition of Na_3AlF_6 lowers the melting point of alumina.
- C) CO_2 is evolved at the anode during electrolysis.
- D) The cathode is a steel vessel with a lining of carbon.

47. Choose the correct statement(s) regarding iproniazid:

- A) It is a neurologically active drug
- B) It inhibits the enzyme which catalyze the degradation of noradrenaline
- C) It is a barbiturate
- D) It is used in controlling depression and hypertension

48. Observe the below three compounds and choose the CORRECT statements about their interconversions

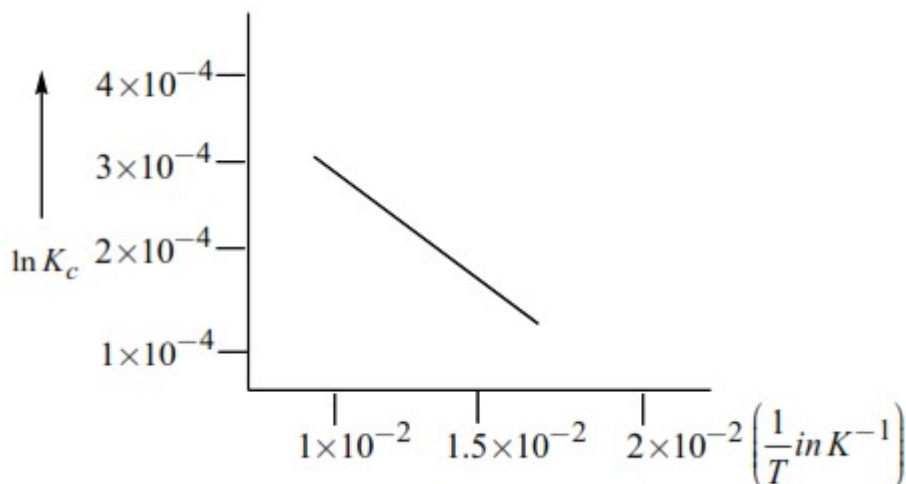
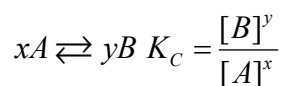
(Consider no reaction between tertiary nitrogen and methyl iodide)



- A) Morphine on treatment with 2 moles of NaH and one mole of methyl iodide gives codeine
- B) Morphine on treatment one equivalent of methyl iodide gives codeine as a major product
- C) Morphine on treatment one equivalent of NaOH and one equivalent of Methyl iodide gives codeine
- D) Morphine on treatment with excess acetic anhydride in basic medium gives heroin

49. A schematic plot of $\ln K_c$ versus inverse of temperature for reaction is shown below

Which of the following statement(s) is/are incorrect regarding the reaction?



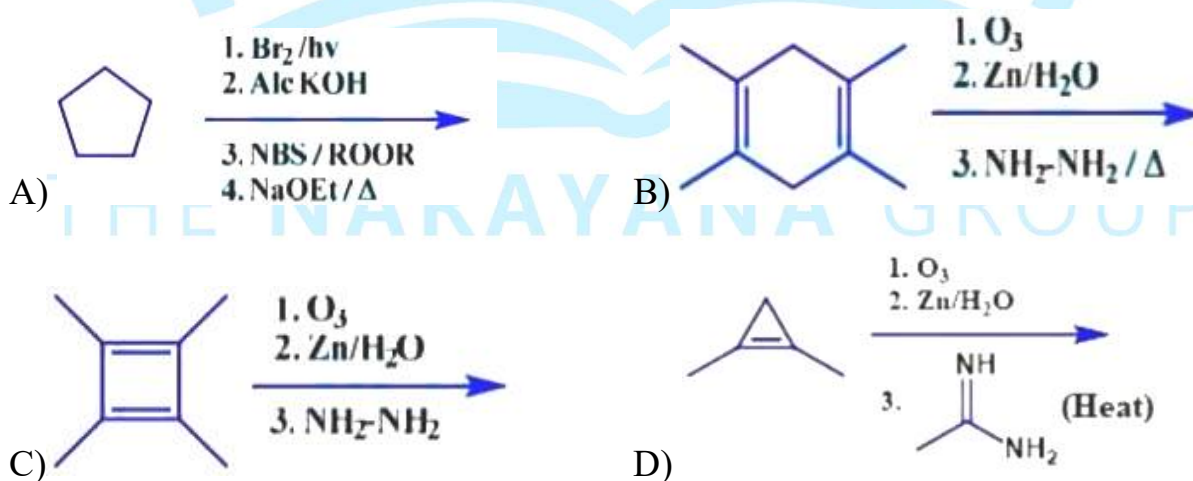
A) Sign of $\Delta_r H^\circ$ is positive

B) Sign of $\Delta_r S^\circ$ is negative

C) $\Delta_r G^\circ$ is negative at all temperature

D) At equilibrium, rate of consumption of A is equal to rate of formation of B

50. Which among the following can yield the aromatic compound



SECTION-III
(Paragraph Type)

This section contains 2 groups of question. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choice (A), (B), (C) AND (D) for its answer, out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph for Questions 51 and 52:

'Equivalent point' in an acid-base titration is defined as that point where equivalents of acid become equal to equivalents of base. In case of polyprotic acids and bases there may be multiple equivalence points. Depending upon the type of substances present, pH of the solution can be calculated at any equivalence point.

Given that :

$$K_{a1}, H_3PO_4 = 5 \times 10^{-4}; (pK_{a1} = 3.3)$$

$$K_{a2}, H_3PO_4 = 5 \times 10^{-8}; (pK_{a2} = 7.3)$$

$$K_{a3}, H_3PO_4 = 4.5 \times 10^{-11}; (pK_{a3} = 10.35)$$

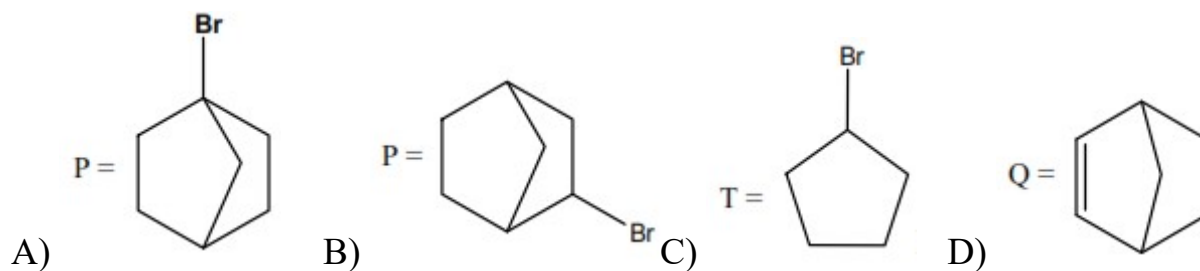
(Use : $\log 2 = 0.3$, $\log 5 = 0.7$; $\log 3 = 0.48$)

51. The pH of the solution when 100 ml of 0.1 M H_3PO_4 is mixed with 200 ml of 0.1 M NaOH solution at 298 K is:
- A) 8.88 B) 1.5×10^{-9} C) 5.3 D) 7.3
52. The pH of the final solution obtained on mixing 200 ml of 0.2 M H_3PO_4 with 50 ml of 0.3 M NaOH solution at 298 K?
- A) 3.3 B) 11.92 C) 3.08 D) 4.16

Paragraph for Questions 53 and 54:

An optically active organic compound P has molecular formula $C_7H_{11}Br$, and it does not decolorize bromine water. P on treatment with potassium tertiary butoxide in tertiary butanol yields Q(C_7H_{10}). The compound Q on oxidation with acidified solution of $KMnO_4$ yields R($C_7H_{10}O_4$), which does not undergo intramolecular dehydration on heating with P_2O_5 . The compound R on heating with NaOH(CaO) gives S(C_5H_{10}). S on heating with dry bromine vapour given T(C_5H_9Br) as only monobromo derivative

53. Select the incorrect structure



54. Major product of the following reaction scheme is

